College Management System

Developed Using Django Framework

Project Name: College
App Name: Department
Model Name: Dept_Details

Name: Mayank Kumar

1. Introduction

The College Management System is a web-based application developed using the Django framework. The system is designed to manage department-related information within a college. It allows the administrator to add, view, and manage department details such as department name, Head of Department (HOD), total staff, and total students.
This project demonstrates the implementation of Django's MVC (Model-View-Template) architecture.

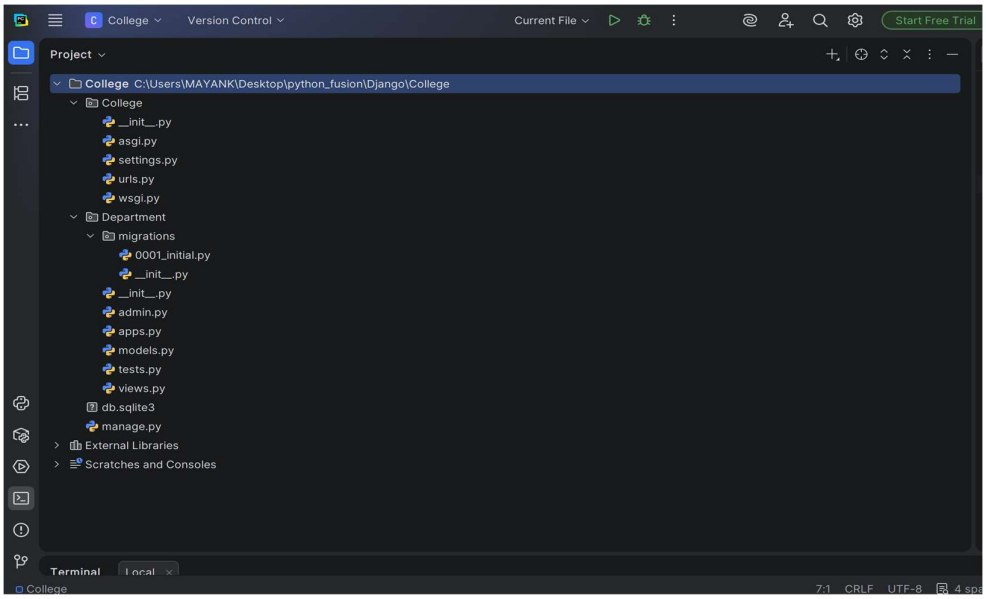
2. Objective of the Project

To develop a web-based College Management System.
To store and manage department details efficiently.
To implement Django Models, Views, Templates, and URL routing.
To integrate Django Admin panel for data management.
To display department data dynamically on the homepage.

3. Technologies Used

- Python 3.14
- Django 6.0.2
- SQLite (Default Django Database)
- PyCharm IDE
- Windows OS

4. Project Structure



5. Steps to create Project.

Step 1: Create Project

Open terminal in project folder and run: **django-admin startproject College**

Move into project directory: **cd ./college**

Check file: ls

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\MAYANK\Desktop\python_fusion\Django> django-admin startproject College
PS C:\Users\MAYANK\Desktop\python_fusion\Django> cd ./College
PS C:\Users\MAYANK\Desktop\python_fusion\Django\College> ls


    Directory: C:\Users\MAYANK\Desktop\python_fusion\Django\College


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        18-02-2026     22:28                College
-a----        18-02-2026     22:28            685 manage.py
```

Step 2: Create App

Open terminal inside project folder and run: **python manage.py startapp Department.**

Move inside project directory: **ls**

College

Department

manage.py

```
PS C:\Users\MAYANK\Desktop\python_fusion\Django\College> python manage.py startapp Department
PS C:\Users\MAYANK\Desktop\python_fusion\Django\College> ls


    Directory: C:\Users\MAYANK\Desktop\python_fusion\Django\College


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        18-02-2026     22:37                College
d-----        18-02-2026     22:37                Department
-a----        18-02-2026     22:28            685 manage.py
```

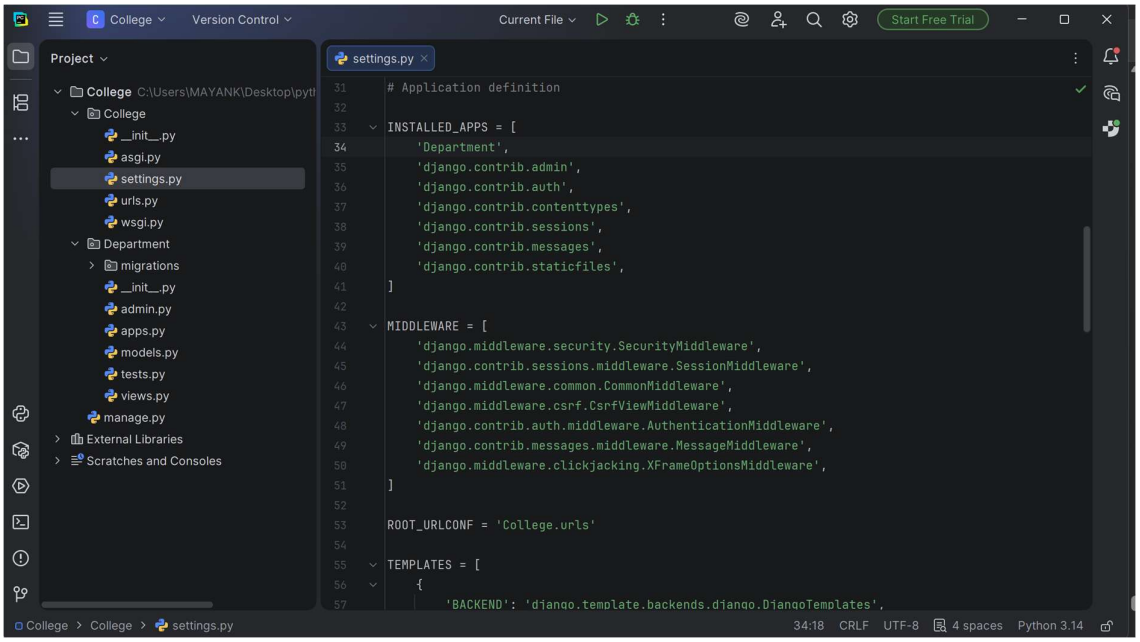
Step 3: Register App in settings.py

Go to: **College → College → settings.py in pycharm.**

Find: **INSTALLED_APPS = []**

Aad application name department in the following list.

Save the file.



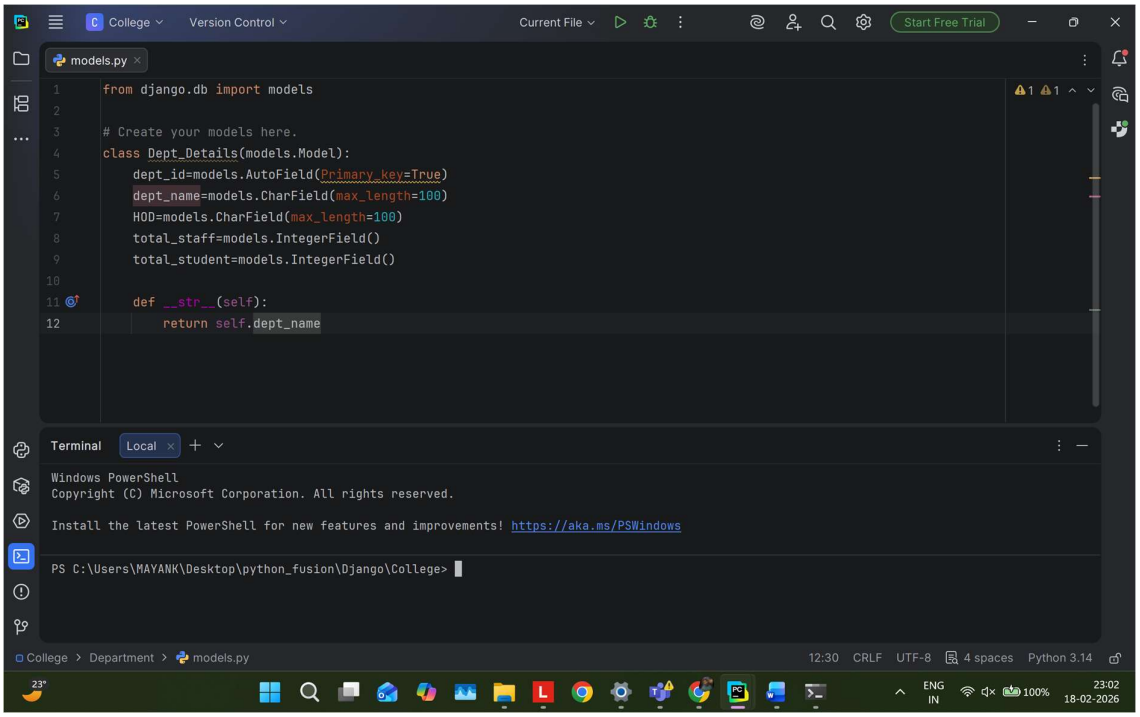
Step 4: Create Model

Go to: **Department → models.py**

Add the following code:

```
from django.db import models
class Dept_Details(models.Model):
    dept_id = models.AutoField(primary_key=True)
    dept_name = models.CharField(max_length=100)
    hod = models.CharField(max_length=100)
    total_staff = models.IntegerField()
    total_student = models.IntegerField()
    def __str__(self):
        return self.dept_name
```

Save the file.

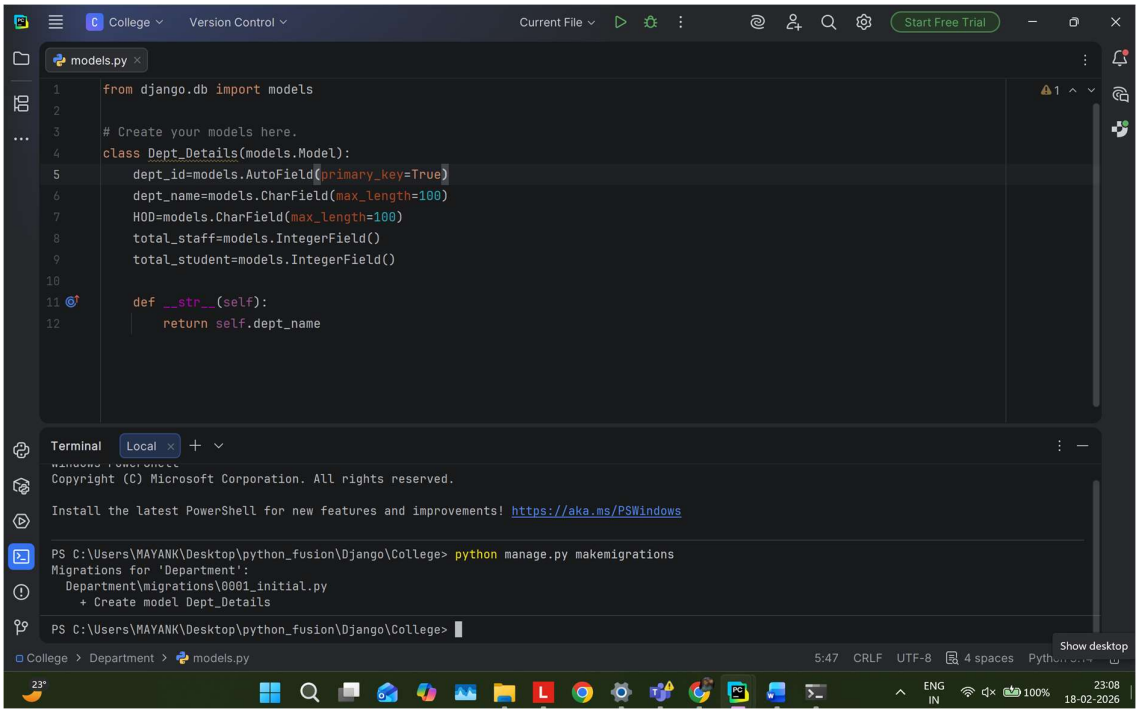


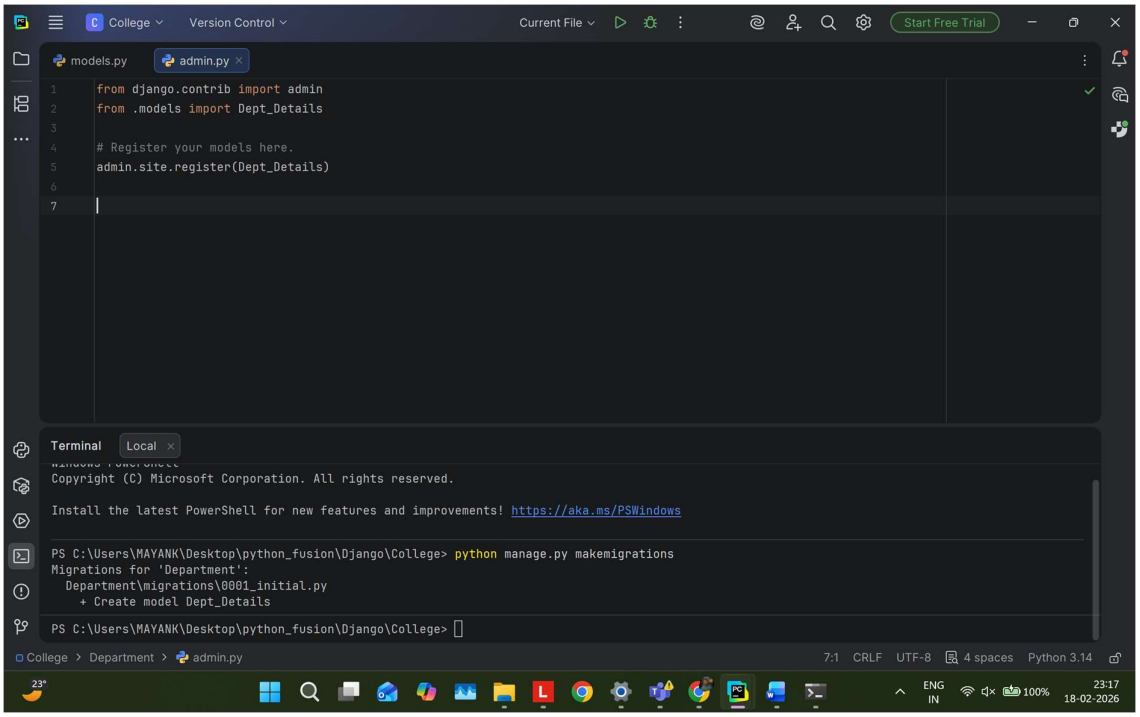
Step 5: Make Migrations

Run: **python manage.py makemigrations**

python manage.py migrate

This creates the database table for Dept_Details.





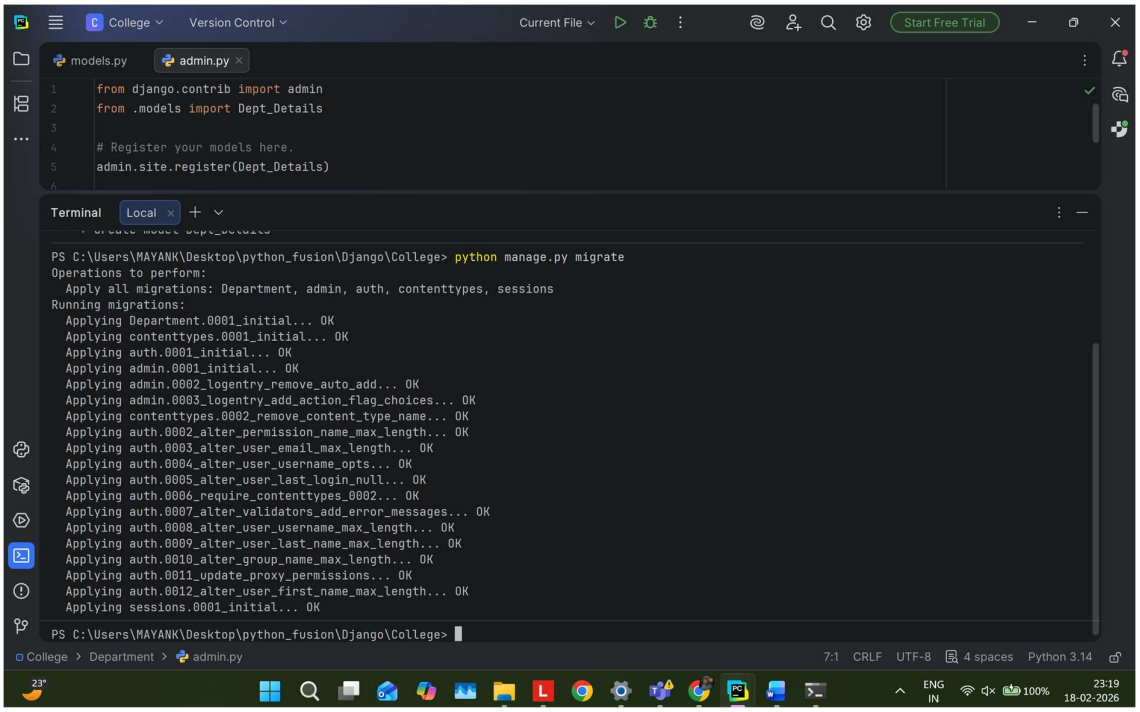
Step 6: Register Model in Admin.

Go to: Department → **admin.py**

Replace the code : **from django.contrib import admin**

from .models import Dept_Details

admin.site.register(Dept_Details)

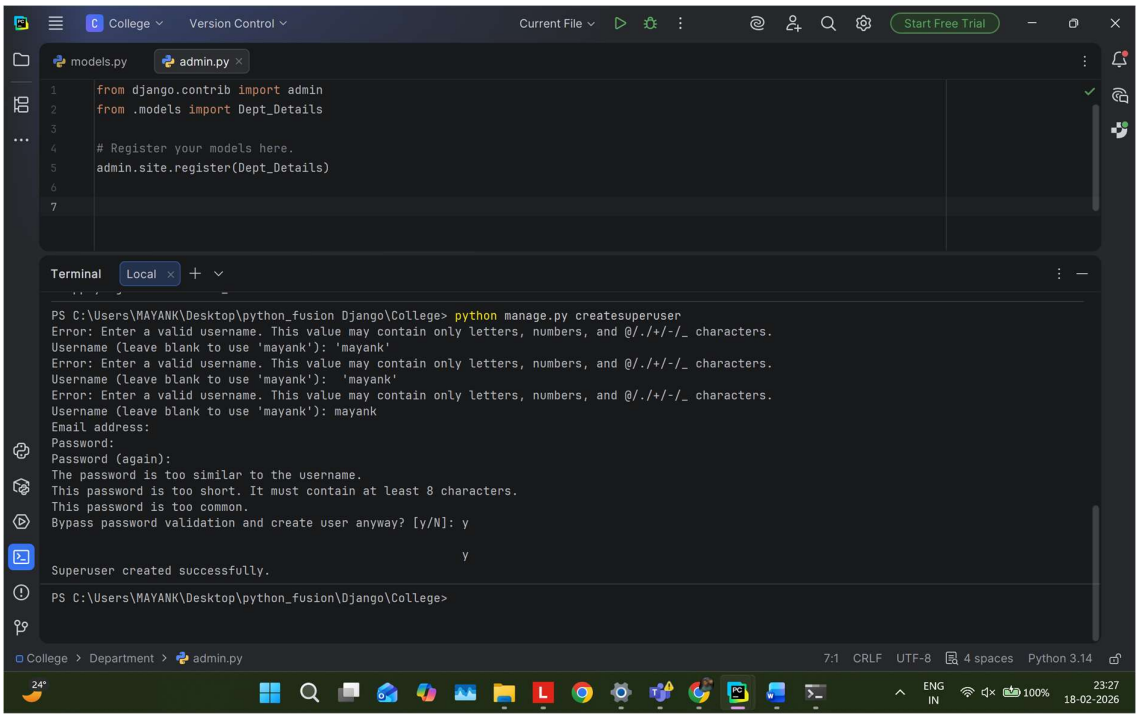

College
Version Control
Current File
Start Free Trial
models.py
admin.py
from django.contrib import admin
from .models import Dept_Details
# Register your models here.
admin.site.register(Dept_Details)
Terminal
Local
PS C:\Users\MAYANK\Desktop\python_fusion\Django\College> python manage.py migrate
Operations to perform:
Apply all migrations: Department, admin, auth, contenttypes, sessions
Running migrations:
Applying Department.0001_initial... OK
Applying contenttypes.0001_initial... OK
Applying auth.0001_initial... OK
Applying admin.0001_initial... OK
Applying admin.0002_logentry_remove_auto_add... OK
Applying admin.0003_logentry_add_action_flag_choices... OK
Applying contenttypes.0002_remove_content_type_name... OK
Applying auth.0002_alter_permission_name_max_length... OK
Applying auth.0003_alter_user_email_max_length... OK
Applying auth.0004_alter_user_username_opts... OK
Applying auth.0005_alter_user_last_login_null... OK
Applying auth.0006_require_contenttypes_0002... OK
Applying auth.0007_alter_validators_add_error_messages... OK
Applying auth.0008_alter_user_username_max_length... OK
Applying auth.0009_alter_user_last_name_max_length... OK
Applying auth.0010_alter_group_name_max_length... OK
Applying auth.0011_update_proxy_permissions... OK
Applying auth.0012_alter_user_first_name_max_length... OK
Applying sessions.0001_initial... OK
PS C:\Users\MAYANK\Desktop\python_fusion\Django\College>
College > Department > admin.py
7:1
CRLF
UTF-8
4 spaces
Python 3.14
ENG
IN
100%
23:19
18-02-2026

Step 7: Create Superuser

Run in pycharm terminal: **python manage.py createsuperuser**

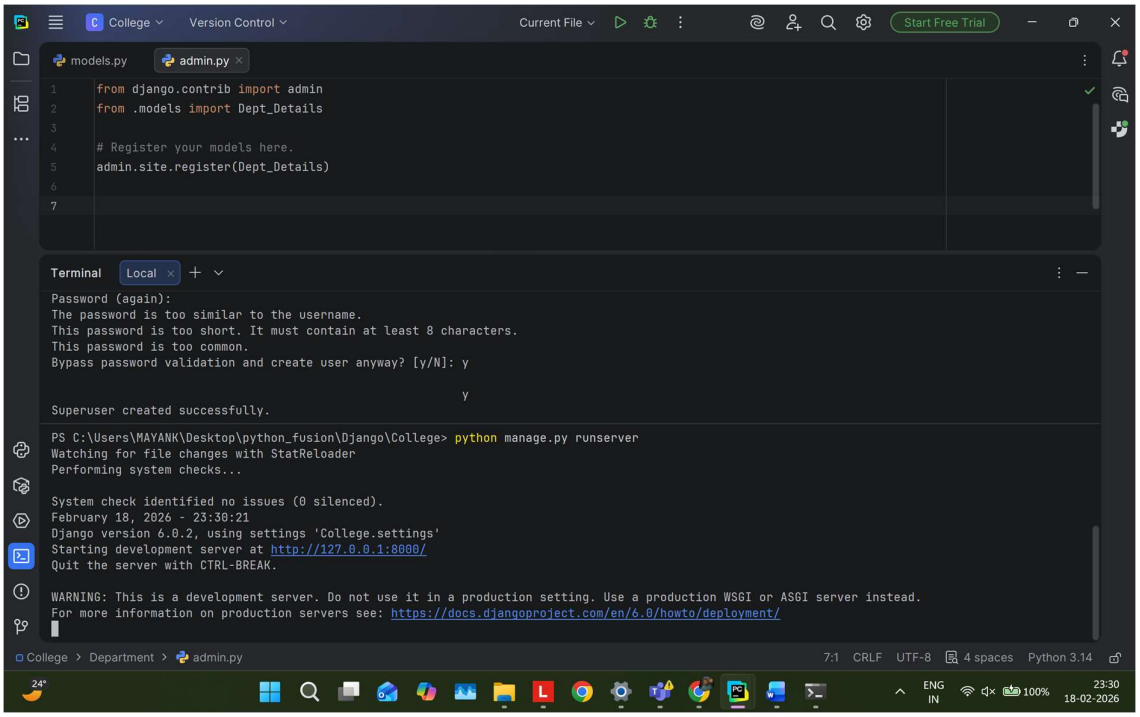
Enter:

- Username
- Email
- Password

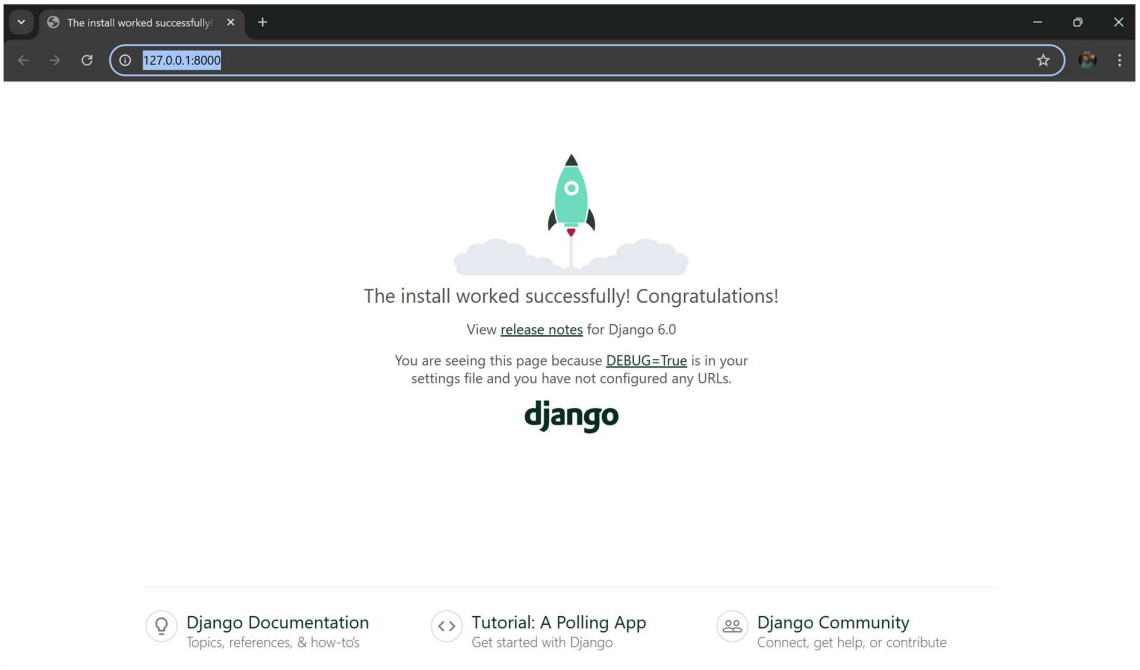


Step 8: Run Server.

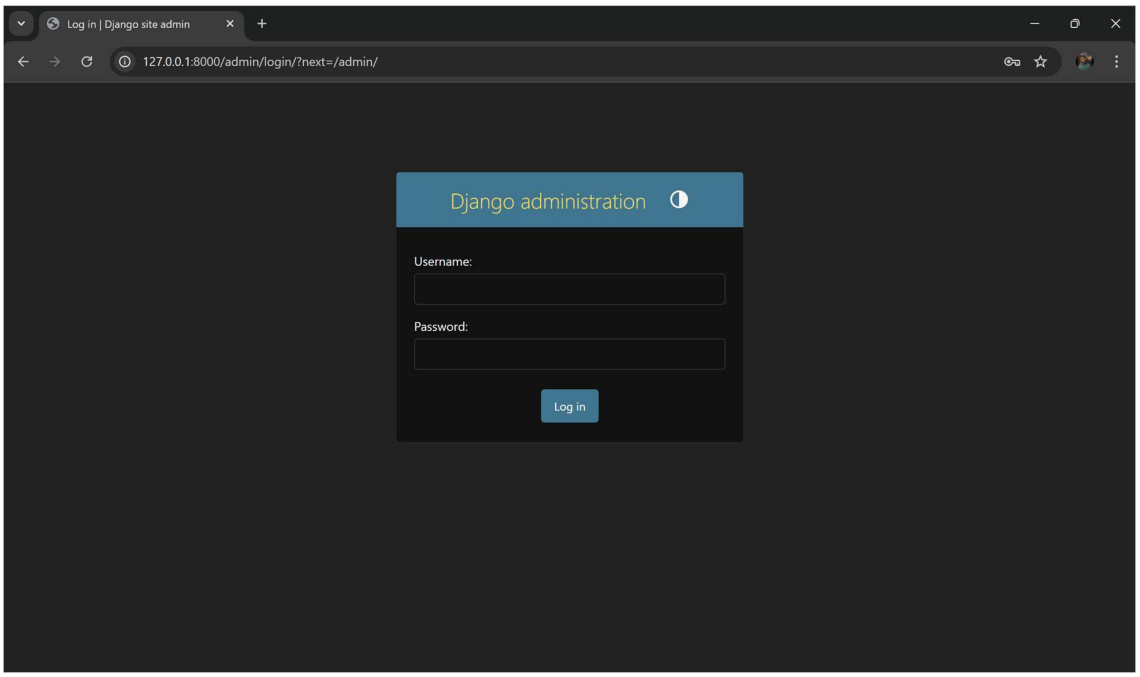
Run in pycharm terminal: **python manage.py runserver**

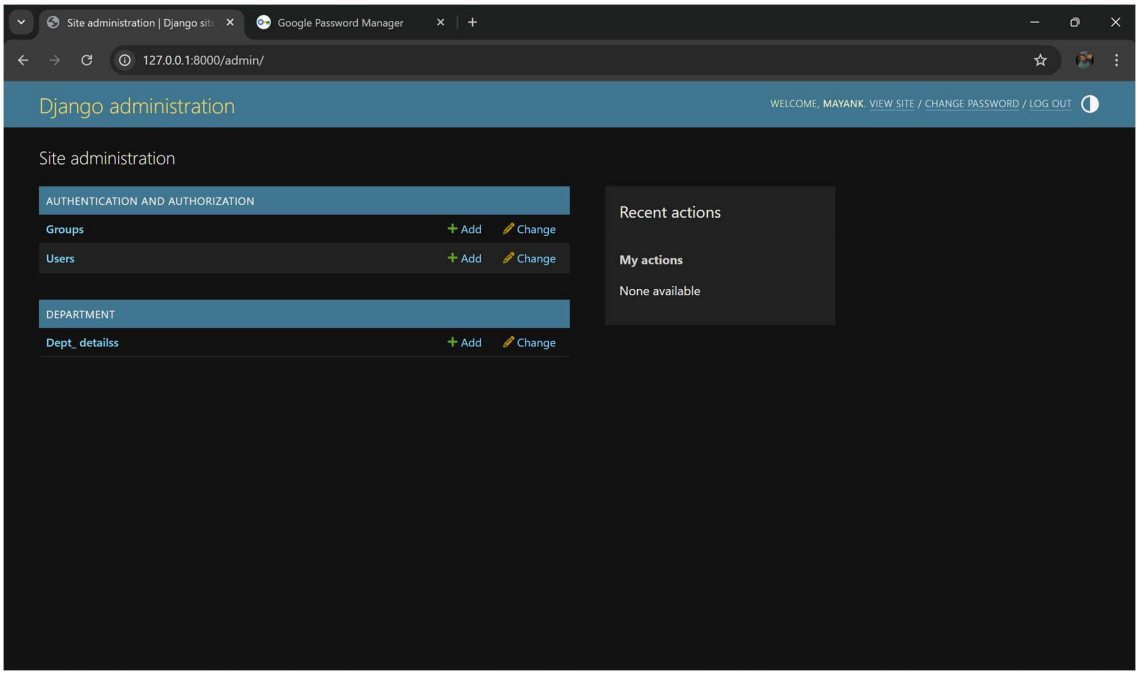


Open browser through link: http://127.0.0.1:8000/

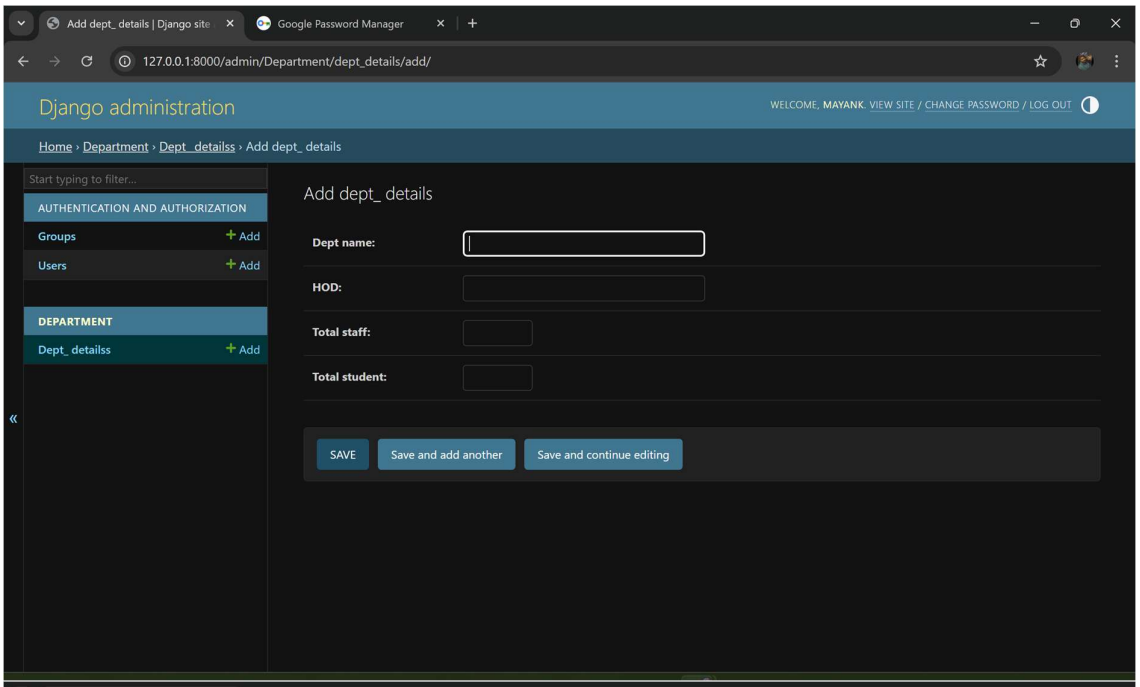


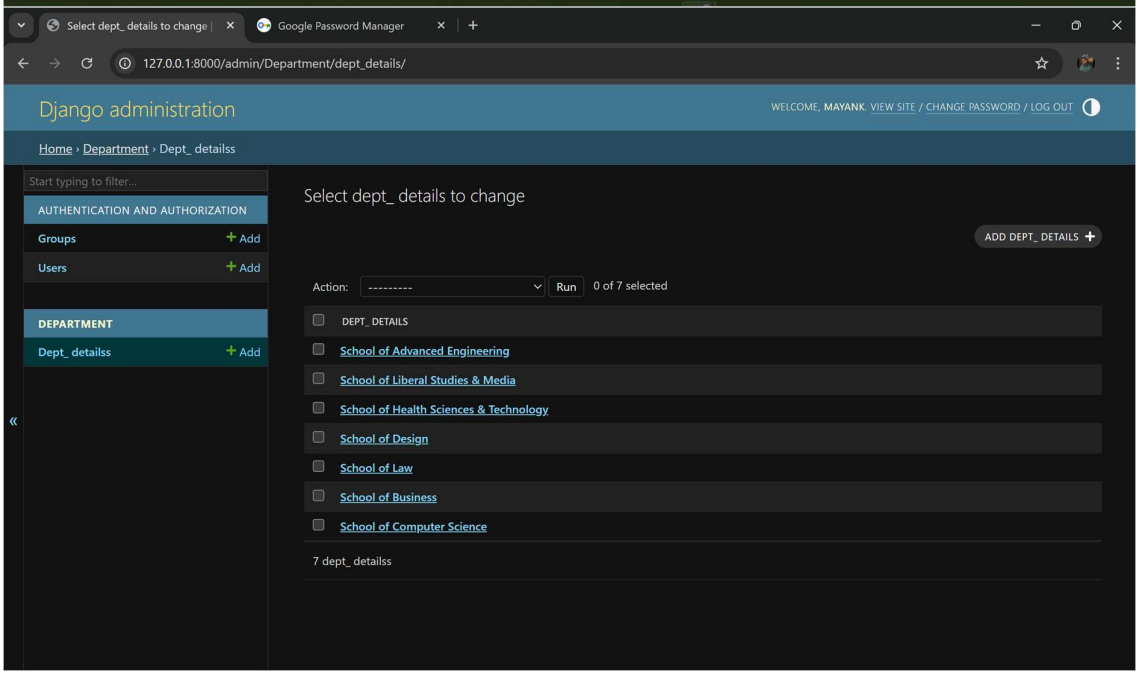
Go to admin page and login through credentials:





Step 9: Aad Details on Admin Panel:





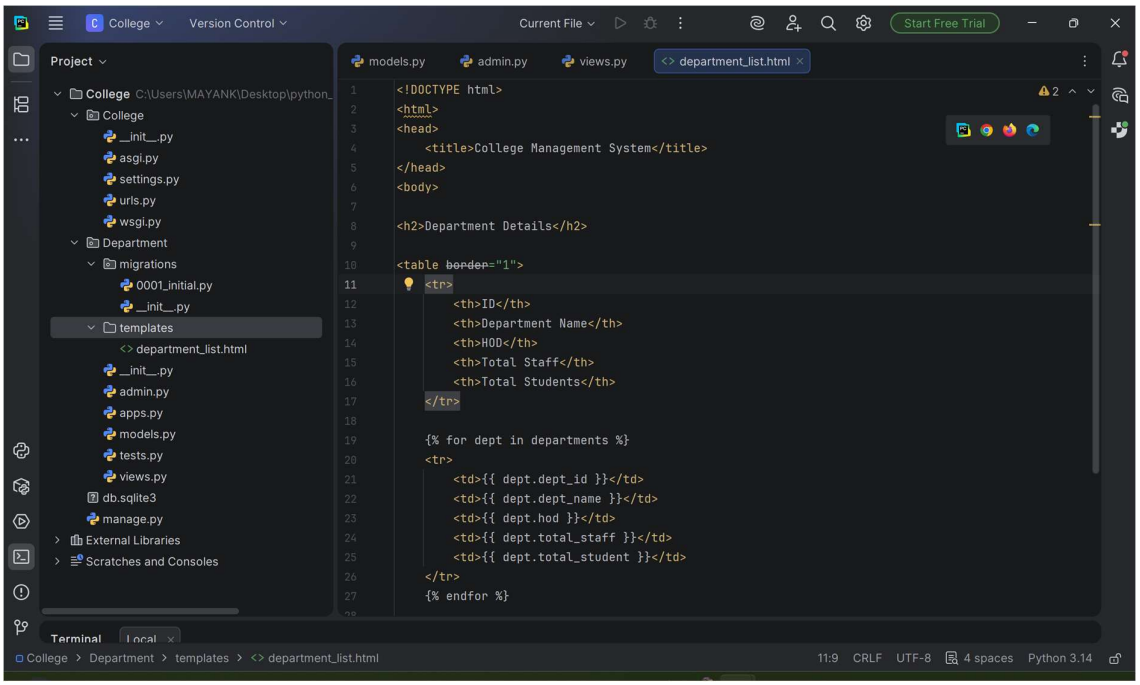
Step 10: Working with template

To display department data on the homepage, we created a template file and connected it with the view.

Go to: **Department → templates → department_list.html**

This template uses Django Template Language (DTL) to dynamically fetch and display data from the database.

Add the following HTML code:



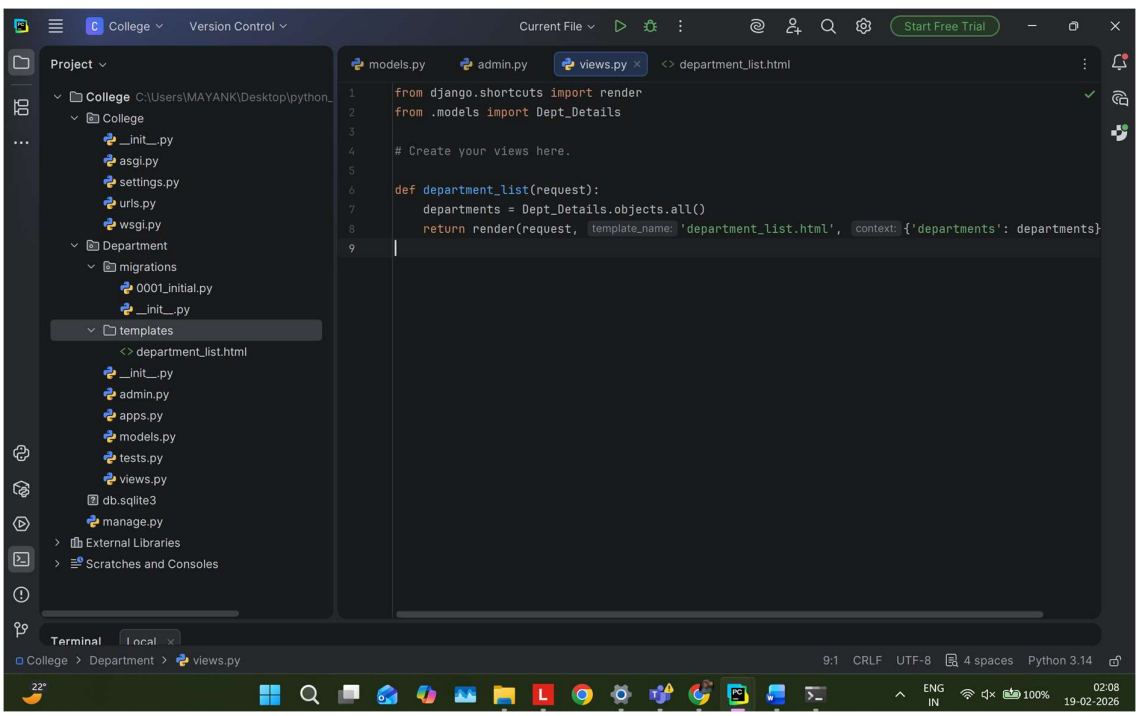
Step 11: Create View

Go to: **Department → views.py**

Add the following code:

```
from django.shortcuts import render
from .models import Dept_Details

def department_list(request):
    departments = Dept_Details.objects.all()
    return render(request, 'department_list.html', {'departments': departments})
```



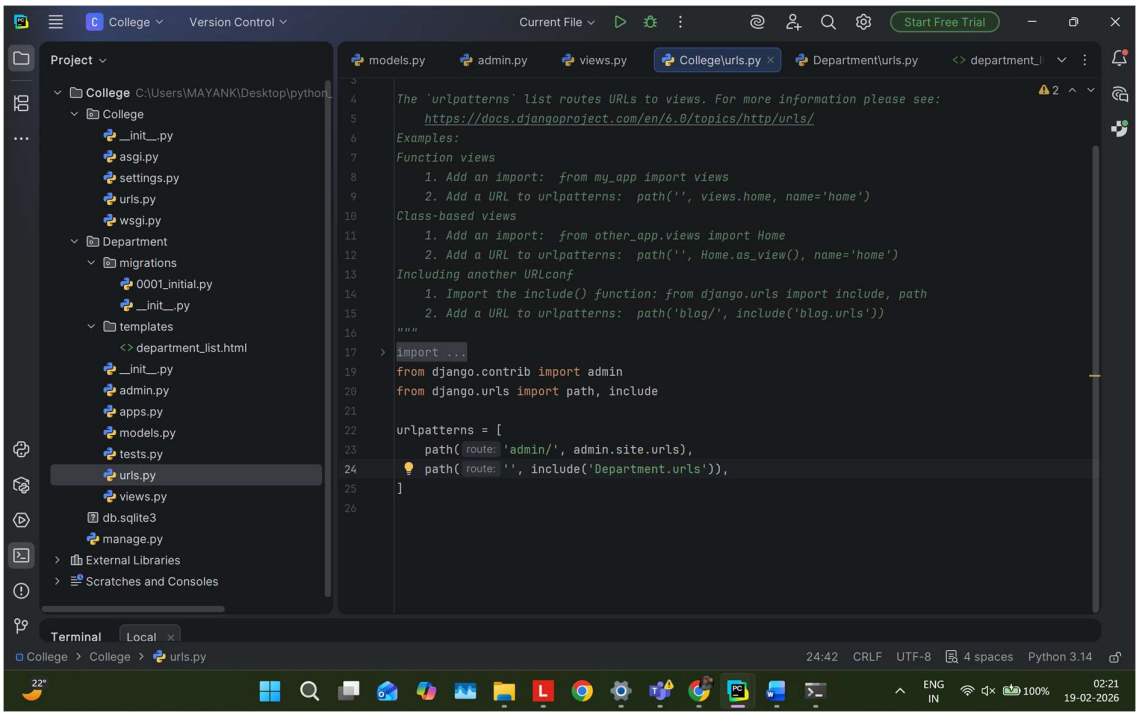
Step 12: Configure URLs

Project Level (College → urls.py)

Add: **from django.urls import path, include**

Update urlpatterns:

```
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('Department.urls')),
]
```



App Level (Department → urls.py)

Create urls.py inside Department folder.

Add:

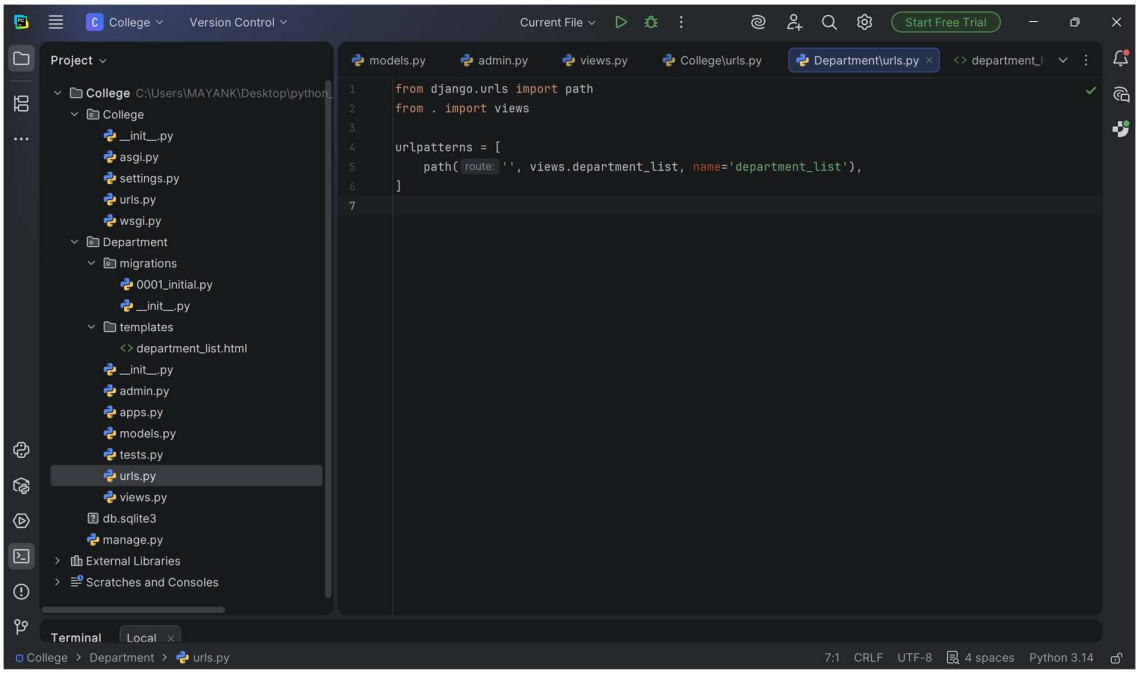
from django.urls import path

from . import views

urlpatterns = [

path('', views.department_list, name='department_list'),

]

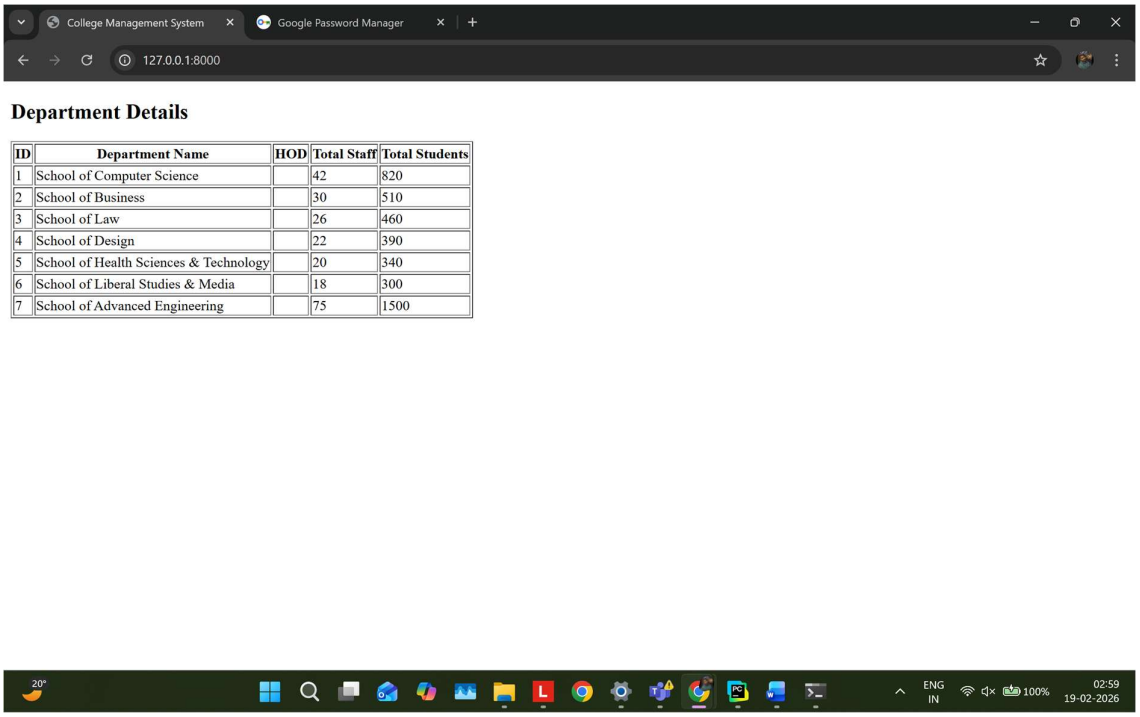


Step 13: Final Output

Run: **python manage.py runserver**

Open browser: http://127.0.0.1:8000/

The homepage will display all department details in table format.



ID	Department Name	HOD	Total Staff	Total Students
1	School of Computer Science		42	820
2	School of Business		30	510
3	School of Law		26	460
4	School of Design		22	390
5	School of Health Sciences & Technology		20	340
6	School of Liberal Studies & Media		18	300
7	School of Advanced Engineering		75	1500